OPENSEC STUDIO V1.10.10 OFFICIAL RELEASE

# Official Illustrated Guide to OpenSec Studio

The Cybersecurity, Web Audit & DevSecOps Suite Explained for Everyone



CREATOR & LEAD SOFTWARE ARCHITECT

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OpenSec Security Engineering

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# 📖 Prologue: The Great Internet Village

> **🏘️ The metaphor of letters and shops**
>
> Imagine that the **Internet** is a bustling village full of houses and shops:
>
> - **Your Web Browser:** You walking down the high street asking for things.
> - **The Web Server:** A shop attended by shopkeepers behind the counter.
> - **An HTTP Request:** A handwritten letter you send saying: *"Please show me the toy catalog"*.
> - **An HTTP Response:** The parcel the shop returns containing what you asked for.

## What is OpenSec Studio?

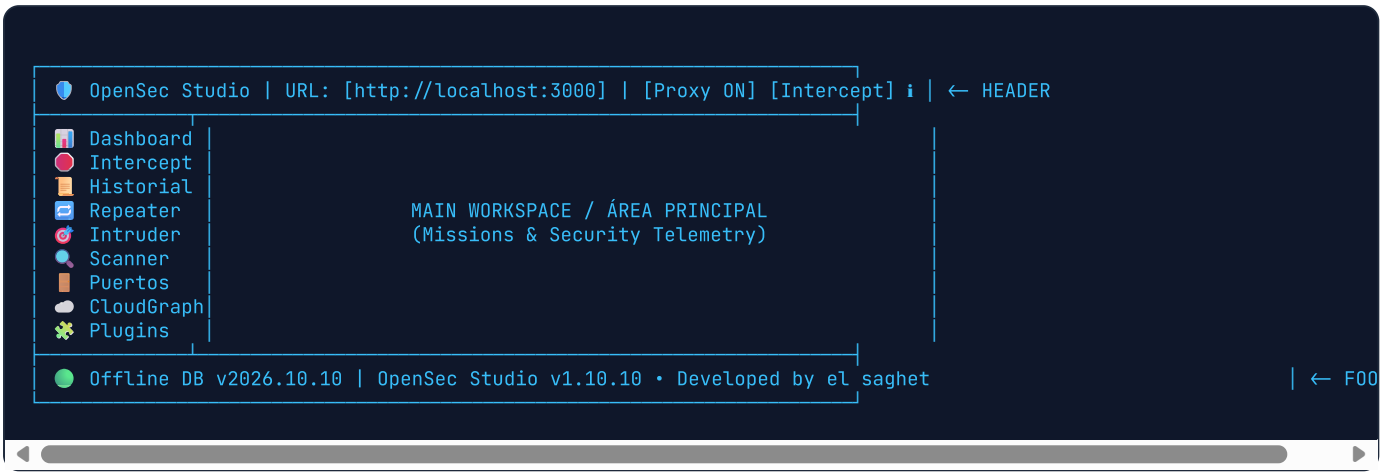
It is your **magic detective backpack** crafted by **el saghet**. Inside it, you carry tools to intercept letters mid-air (Interceptor), duplicate and send them a thousand times (Repeater), test key combinations (Intruder), check castle gates (DAK Scanner & Port Scanner), audit server configurations (SSH Host Audit), and consult a secret encyclopedia with over 1,200 threats even on airplane mode (Offline Feed).

> **🌟 Fundamental Principle: 100% Real, Zero Mock Data**
>
> In OpenSec Studio no button is a placeholder or mock simulation. Every single action executes native Rust code, communicates over live network sockets, and processes authentic packets.

# Chapter 1: First Flight & Screen Anatomy

Go to your Windows Desktop or root folder and double-click `OpenSec Studio.exe`. The window opens instantly, hardware-accelerated and in focus.

## Visual Interface Anatomy:




| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **1. Header (Top Bar)** | Quick controls: Target URL, Proxy ON/OFF toggle, Intercept state, and About info (ℹ️). | The steering wheel and dashboard switches. |
| **2. Sidebar (Navigation)** | Instant navigation across all 18 cybersecurity modules. | Your organized tool rack. |
| **3. Central Workspace** | Displays code editors, traffic grids, metrics, charts, and audit findings. | Your detective workbench and lab. |

### 🕵️ Detective Mission 1: The Official Badge

Click the ℹ️ button in the top right corner. The official modal will display version **v1.10.10** and credits to **el saghet**.

# Chapter 2: The Dashboard Control Panel

The Dashboard brings together all live audit telemetry in real-time.

> 🚗 **Speedometer and Gauges**
>
> Just like the cockpit of a sports car, it indicates network speed, engine status, and immediate road hazards.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Total Requests** | Cumulative count of inspected network traffic. | Total miles driven. |
| **Critical Vulnerabilities** | Red flashing alert for severe flaws needing immediate fixes. | Check-engine warning light. |
| **Security Score** | Global percentage security posture of the target. | Vehicle overall health status. |

> 🕵️ **Detective Mission 2: Engage the Radar**
>
> Click the primary Proxy switch. Notice the status transition from "Paused" to vibrant green "Active".

# Chapter 3: The Customs Officer (HTTP Interceptor & HTTPS Tunnels)

The Interceptor sits between your browser and the remote server to inspect packets before transmission.

> 🔴 **The customs inspector with a clipboard**
>
> Before a letter travels, the guard catches it mid-air. You can read it, edit contents with a pencil, and decide whether to Forward or Drop it.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Intercept is ON / OFF** | Enables or pauses the mandatory packet inspection halt. | Lowering or raising the tollgate barrier. |
| **Forward (Proceed)** | Sends the edited letter onwards to the destination server. | Waving the inspected vehicle through. |
| **Drop (Discard)** | Terminates transmission and closes the socket connection. | Confiscating and destroying suspect cargo. |

> 🕵️ **Detective Mission 3: Edit a Message in Mid-Air**
>
> Enable interception, submit a form in your browser, alter a field value inside OpenSec, and click Forward.

# Chapter 4: The Captain's Logbook (HTTP History)

Immutably captures every single request and response processed through OpenSec Studio.

> 📜 **The royal archivist ledger**
>
> Every letter sent is logged with timestamp, HTTP method, URL, payload size, and color-coded status codes.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **200 OK (Green)** | The shop fulfilled the order smoothly. | Successful package delivery. |
| **404 / 403 (Amber)** | Resource missing or door locked. | Missing address or restricted room. |
| **500 Error (Red)** | Server suffered an internal failure. | Kitchen fire in the shop backroom. |

> 🕵️ **Detective Mission 4: Filter Sensitive Traffic**
>
> Type "/api" in the History filter bar to immediately isolate backend web service endpoints.

# Chapter 5: The Magic Photocopier (Native Rust Repeater)

Dispatches authentic HTTP requests via native Rust sockets with millisecond latency calculation.

> 🔁 **The instant cloning machine**
>
> Instead of retyping a web form 100 times, the copier allows modifying a single header or parameter and firing it instantly.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Multi-Tab Workspace** | Enables concurrent work on multiple endpoints. | Multiple open sheets on your desk. |
| **Send Button** | Fires the raw request over the native Rust engine. | Pushing the copier start button. |
| **Split-View Viewer** | Displays request on the left and response on the right. | Side-by-side letter comparison. |

> 🕵️ **Detective Mission 5: Hidden Header Inspection**
>
> Send a request to your test endpoint and observe the exact millisecond round-trip time in the response tab.

# Chapter 6: The Key-Testing Robot (Intruder)

Automated fuzzing and dictionary brute-force engine capable of high-throughput concurrent payload delivery.

> **The high-speed locksmith automaton**
>
> Given a padlock and a sack of 1,000 keys, the robot tests every key in split seconds and flags which one unlocked the door.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Markers §payload§** | Marks the injection positions in the request template. | Drawing a red cross where the key inserts. |
| **Sniper Mode** | Tests the wordlist against one position at a time. | A sharpshooter targeting one spot. |
| **Cluster Bomb Mode** | Tests all combinations of usernames and passwords. | Testing every key against every lock. |

**Detective Mission 6: Discover Hidden Endpoints**

Configure a wordlist attack with paths (admin, backup, login) and look for differing response status codes.

# Chapter 7: The Live Security Inspector (DAST Scanner & YAML Templates)

Dynamic application vulnerability scanning detecting OWASP Top 10, SQLi, and Cross-Site Scripting.

> 🔍 **The bloodhound with a magnifying glass**
>
> The inspector checks every wall, window, and padlock searching for vulnerabilities the builders forgot to secure.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Passive Scan** | Analyzes headers and responses without aggressive injection. | Inspecting from the sidewalk without touching. |
| **Active Scan** | Sends targeted security payloads to confirm vulnerabilities. | Gently rattling the window to check its lock. |
| **SARIF & HTML Export** | Generates OASIS-compliant reports for engineering teams. | The signed official detective verdict. |

> 🕵️ **Detective Mission 7: Security Headers Audit**
>
> Launch a passive scan against localhost and notice missing Content-Security-Policy (CSP) detections.

# Chapter 8: The Door Knocker & Locksmith (Port Scanner & SSH Audit)

Scans all 65,535 TCP ports with asynchronous sockets and executes credentialed SSH host compliance checks.

> 🚪 **Knocking on every door in the palace**
>
> Verifies which doors are open (80 HTTP, 443 HTTPS, 22 SSH) and identifies which service guards each gate.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Concurrent TCP Scan** | High-speed port probe written in native Rust. | A team of messengers ringing bells together. |
| **SSH Host Audit** | Evaluates Linux/Windows configurations against CIS Benchmarks. | Auditing the inner steel vault fortifications. |

> 🕵️ **Detective Mission 8: Discover Open Web Ports**
>
> Run a port scan against 127.0.0.1 for ports 80–8088 and watch live discovered services populate.

# Chapter 9: The Magic Recipe Book (Nuclei-Style YAML Templates)

Create custom vulnerability detection rules using industry-standard declarative YAML syntax.

> 📜 **The book of magical recipes**
>
> Each template is a cooking recipe: it specifies what request to send and what string confirms a flaw.

| Section | Function in OpenSec Studio | Metaphor to understand it |
| --- | --- | --- |
| **Syntax-Highlighted Editor** | Live template editing with real-time parser validation. | Writing spells with enchanted ink. |
| **Matchers & Extractors** | Pattern matching rules using strings or regular expressions. | The secret ingredient revealing the trap. |

> 🕵️ **Detective Mission 9: Build a Custom Rule**
>
> Create a YAML template to detect exposed /.git/HEAD files and validate it in the live syntax checker.

# Chapter 10: The Universal Translator (Decoder)

Transforms text and payloads across multiple encodings: Base64, URL Encode, Hex, HTML entities, and Hashes.

> 🔑 **The detective Rosetta stone**
>
> Spies often encode messages so others cannot read them. The translator renders them instantly understandable.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **URL Encoding (%20)** | Encodes spaces and special characters for web safety. | Packaging words safely for postal transit. |
| **Base64 & Hex** | Translates binary data into readable character strings. | Translating secret glyphs into known letters. |
| **SHA-256 / MD5 Hashes** | Generates irreversible cryptographic fingerprints. | The unique biometric fingerprint of a letter. |

> 🕵️ **Detective Mission 10: Decode a Base64 Token**
>
> Paste a Base64 string into the input pane, choose "Decode as Base64", and inspect the decoded cleartext output.

# Chapter 11: The Building Materials Inspector (Live SCA Audit)

Analyzes software dependency manifests (package.json) to detect outdated or compromised packages.

> 🧱 **The brick and beam inspector**
>
> Checks whether building materials have known flaws or factory recalls.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Library CVE Correlation** | Cross-references package versions with CVE databases. | Consulting the catalog of recalled materials. |
| **Remediation Advice** | Identifies the minimum secure version required. | The architect order to replace a cracked beam. |

> 🕵️ **Detective Mission 11: Audit a Dependency Manifest**
>
> Load a package.json file into the SCA module and inspect the list of flagged component vulnerabilities.

# Chapter 12: The Treasure Map (SiteMap & JS Spider)

Builds a hierarchical tree of target endpoints through passive traffic inspection and client JavaScript parsing.

> 🗺️ **The cartographer map of the realm**
>
> As you navigate, the cartographer charts every alleyway, storefront, and secret door discovered.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Hierarchical Tree** | Groups endpoints by host, directory, and file name. | Branches of a family tree. |
| **Passive JS Route Extractor** | Discovers hidden API routes inside frontend JavaScript scripts. | Uncovering secret notes written by the builders. |

> 🕵️ **Detective Mission 12: Explore Server Structure**
>
> Browse through a web app with the proxy on and observe how the SiteMap automatically constructs the directory layout.

# Chapter 13: The Spy Postman (OAST Collaborator Client)

Out-of-Band testing client detecting blind vulnerabilities (Blind SSRF, Log4Shell, Out-of-Band SQLi).

> 🔔 **The trap with a secret bell**
>
> You leave a letter containing a secret return address. If the server accesses it, the bell rings and you confirm the flaw.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Unique Domain Generator** | Generates correlation IDs for payload tracking. | The secret serial number on the trap. |
| **DNS & HTTP Interaction Log** | Shows when and from which IP the server visited the payload. | The bell alerting you when the trap was sprung. |

> 🕵️ **Detective Mission 13: Deploy an OAST Trap**
>
> Generate a collaborator payload and watch incoming DNS/HTTP interactions appear in real-time.

# Chapter 14: The Time Machine & the Judge (Diffing & PCI-DSS Audit)

Compares previous audit scans with new runs to track fixed versus persistent flaws and calculates regulatory compliance.

> ⚖️ **The scales of justice and the time capsule**
>
> Compares yesterday photograph with today to reward fixed flaws and flag new regressions.

| Section | Function in OpenSec Studio | Metaphor to understand it |
| --- | --- | --- |
| **Vulnerability Diffing** | Categorizes flaws as Fixed, New, or Persistent. | The checklist of completed and outstanding tasks. |
| **PCI-DSS v4.0 Compliance** | Computes overall banking industry security adherence. | The official test score from the regulatory judge. |

> 🕵️ **Detective Mission 14: Compare Two Scans**
>
> Load two prior scan files and inspect how the diffing engine highlights resolved vulnerabilities in green.

# Chapter 15: The Great Secret Encyclopedia (Offline CVE Feeds)

Offline database of 1,200+ threats searchable in under 5 ms without requiring any active internet connection.

> 📖 **The secret encyclopedia in your backpack**
>
> A comprehensive handbook of digital villain tactics accessible inside air-gapped bunkers or during airplane flights.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Sub-5ms Search Engine** | Optimized in-memory indexing for instantaneous querying. | Opening the book directly to the target page. |
| **Automatic Delta Sync** | Updates new CVE definitions when connected to the internet. | The courier delivering new volumes to your library. |

> 🕵️ **Detective Mission 15: Search for a Critical Threat**
>
> Search for "Log4Shell" or "Spring4Shell" to read attack vectors and remediation code snippets.

# Chapter 16: The Secret Mission Safe (.opensec Projects)

Saves and restores complete investigations encrypted with AES-256-GCM and PBKDF2 key derivation.

> 🔒 **The armored detective lockbox**
>
> Locks all your traffic history, notes, and Repeater tabs with password protection so no adversary can spy on your work.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **Save Project (.opensec)** | Serializes the entire workspace into an encrypted file. | Locking all evidence inside the vault. |
| **AES-256 Encryption** | Guarantees confidentiality for enterprise audit data. | The unbreakable combination lock. |

> 🕵️ **Detective Mission 16: Encrypt Your Investigation**
>
> Select "Save Project (.opensec)" from the top menu, assign a master passphrase, and inspect the created file.

# Chapter 17: The Extension & Tool Store (BApp Community Store)

Modular extension ecosystem allowing community security plugins to be installed or created in JSON format.

> 🧩 **The detective accessory shop**
>
> Where you equip your backpack with new gadgets: special flashlights, decoders, or radar sensors.

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **1-Click Install** | Dynamically enables or disables plugins. | Snapping an accessory onto your utility belt. |
| **Load Local Plugins** | Imports custom JSON extensions developed by your team. | Crafting your own gadget in the workshop. |

> 🕵️ **Detective Mission 17: Browse the Plugin Store**
>
> Navigate to the Plugins tab, review available community tools, and enable an extension to enhance your workflow.

# Chapter 18: SSL Certificates, 60-Day Trial, $30 BTC Donation & Credits

Root X.509 certificate management for HTTPS interception, 60-day continuous evaluation, and Bitcoin donation radar.

> 🛡️ **The diplomatic trust passport and honor plaque**
>
> The certificate is your diplomatic passport telling the browser: "Trust this officer, he is here to defend you".

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **CA Root Certificate** | Generates and installs trust certificates to inspect HTTPS without warnings. | The detective official wax seal. |
| **60-Day Evaluation Period** | 60 days of full unrestricted access protected by deep multi-anchor persistence. | Your two-month courtesy pass to explore the realm. |
| **Bitcoin $30 USD Donation** | Support independent software with live mempool payment detection. | Your contribution to the digital detectives guild. |
| **Authorship & Credits** | Designed and crafted with pride by el saghet. | The artisan mark engraved into the tool. |

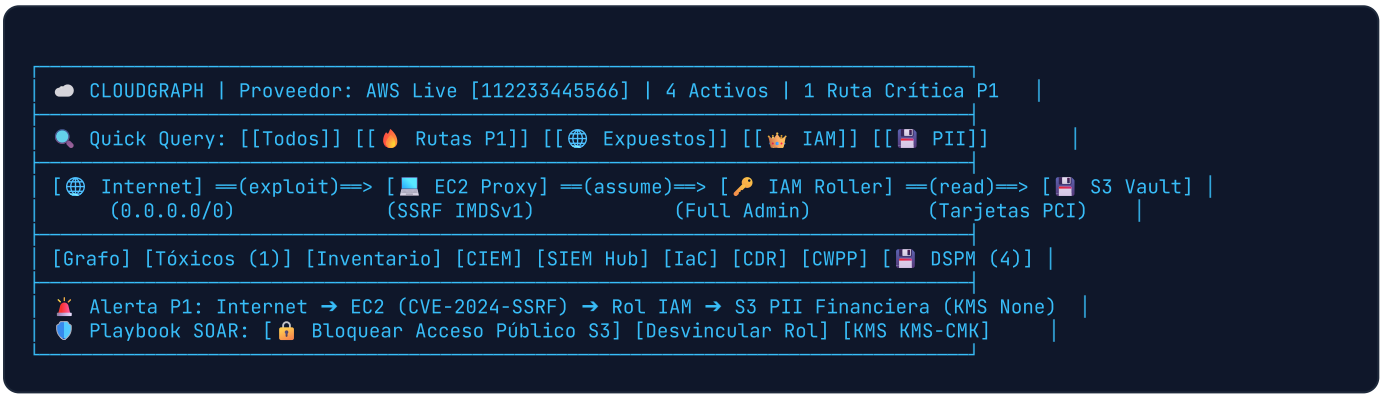
> 🕵️ **Detective Mission 18: The Official Honor Badge**
>
> Open the About OpenSec Studio modal and view the official badge: Developed by el saghet • Version v1.10.10 Official 2026.

# Chapter 19: The Cloud Star Map (CloudGraph: CNAPP, Wiz Engine, DSPM & SIEM Hub)

Comprehensive cloud security posture auditing (CSPM, CIEM, CWPP, CDR, DSPM & IaC) inspired by Wiz architecture. Visualizes the cybersecurity graph, correlates toxic combinations, and connects to LocalStack emulators and open-source vulnerable labs in 1-click.

> 🌌 **The floating cloud kingdom and the toxic combination radar**
>
> Imagine your organization owns castles floating across different clouds (AWS, Azure, GCP). CloudGraph is a 3D radar that maps every bridge, vault, and gateway. Instead of overwhelming you with thousands of isolated alerts, it pinpoints the "perfect storm": an exposed gateway leading to a vulnerable server holding admin keys into the sensitive database.

## Topología y Grafo de Ataque Multidimensional (Wiz-like CNAPP):




| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **CyberGraph & Quick Queries** | Interactive attack graph canvas with instant filter chips: [🔥 Critical P1 Paths], [🌐 Exposed + CVEs], [👑 Admin IAM], and [💾 PII Stores]. | The thermal telescope highlighting only the real infiltration corridors. |
| **Prioritized Toxic Combinations** | Wiz Core correlation engine linking internet exposure, exploitable CVEs, excessive IAM permissions, and sensitive data. | The anatomy of a heist: open door + broken lock + skeleton key + gold vault. |
| **CIEM & IAM Privilege Escalation** | Detection of overprivileged identities, wildcard permissions, and unused credentials. | Finding out the castle apprentice holds the master key to the royal treasury. |
| **DSPM & Sensitive Data Posture** | Deep classification of PCI-DSS credit cards, PII/SSN, and HIPAA records across S3, Azure Blob, and RDS with 1-click KMS SOAR playbooks. | The customs scanner identifying gold vaults and enforcing enchanted KMS locks. |
| **CWPP & Container Security** | Agentless inspection of container registries (ECR, ACR, GAR), hardcoded secret leaks, and admission control quarantine. | Inspecting maritime shipping crates before loading and confiscating contaminated cargo. |
| **CDR & MITRE ATT&CK for Cloud** | Real-time heuristic threat detection across audit logs (CloudTrail and Azure Activity). | Infrared motion sensors catching intruders sneaking through the perimeter at night. |

| Section | Function in OpenSec Studio | Metaphor to understand it |
|---|---|---|
| **IaC Shift-Left Audit** | Pre-deployment security review of Terraform plans (JSON) and Kubernetes manifests (YAML). | Inspecting the blueprint and fixing architectural flaws before pouring concrete. |
| **Local Emulators & Labs (Options 2 & 4)** | Native connection to LocalStack (localhost:4566), Minikube/KinD, and 1-click loading of OWASP CloudGoat, Terragoat, and DVKS. | The military obstacle course where agents train for free without spending live ammunition. |
| **SIEM Hub (Sentinel / Splunk)** | Automated dispatch of toxic findings and analytical query generation in Sentinel KQL and Splunk SPL. | The direct red hotline to law enforcement headquarters. |

**🕵️ Detective Mission 19: Audit a Vulnerable CloudLab in 1-Click**

Open CloudGraph, click "Connect Cloud / IaC", switch to "Vulnerable Labs", hit "⚡ Load Test Data / Lab", and watch the P1 toxic attack path light up in the CyberGraph.